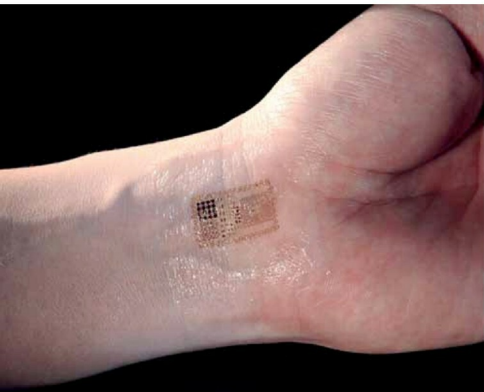


Portable Medical Electronics: What's Inside?

Making smarter portable devices calls for more capable MCUs. Semiconductor manufacturers are rising to the challenge by coming up with innovative solutions that make medical devices portable, low-power, safe, secure, reliable and connected



Electronic patch which bends, wrinkles and stretches like skin, can contain electronic components for sensing, communication and relaying information from the body to a machine (Image courtesy: www.smartplanet.com)

UMA GUPTA

Portable medical devices are improving the healthcare for millions of people. Sophisticated yet affordable, these devices let users monitor their vital signs themselves at home and on the go. Products such as blood glucose monitors, heart rate monitors, ingestible gastrointestinal tract monitors and pain-blocking implants have enhanced the quality of life for those with chronic or acute diseases

and conditions.

The coming years will continue to see many new products powered by innovative semiconductor devices.

Benefits of going small

While the trend towards miniaturisation is fueled to a large extent by the drive to reduce cost by leveraging the cost advantages generated by Moore's Law, there are other factors too at play. The most significant among these is the increased focus by semiconductor

suppliers on the medical area. Many of them are coming out with more and more solutions tailored to particular applications. The longer-term rationale is that by enabling miniaturisation with more tailored solutions, new markets and users will be created. This will increase volumes and thus financial return on the investment.

A case in point is ultrasound. Traditionally, ultrasound machine was a cart-based unit. More recently, hand-held devices have become available with the market no longer limited to hospital use. These devices are little bigger than a mobile phone, packing higher-density measurement electronics such as the AD9278 octal-channel ultrasound receiver from Analog Devices. These miniaturised ultrasound devices have the potential to find their way into the hands of every medical doctor, making ultrasound diagnostics a routine procedure in clinics. One day, they could even replace the stethoscope as the main diagnostic tool of the profession.

Another important driver is the emergence of open mobile phone platform. We are starting to see devices such as glucose meters, heart rate monitors and even pulse oximeters based on mobile phone platforms. The decision to throw open the iPhone and Android platforms has brought about a revolution in medical and healthcare applications. The ultra-low cost and easy availability of mobile phones are compelling arguments in favour of their use in medical field.

One concern, though, is the level of control over such applications. Regula-